

A Review of Machine Learning Methods Applied in Sunspot Prediction

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Abstract—Solar activity has a significant impact on the production and life of human beings and organisms on earth. Since solar activity varies in a cycle of approximately 11 years, the variation of ionospheric plasma density, or of the sunspots has major effect in the space around Earth in the interplanetary environment and the magnetosphere of the Earth. Therefore, information on future changes in the number of sunspots is important for space weather safety. This paper aims to review recently innovate methods of Machine learning (ML) applied in forecasting sunspot number of near-future in the literature. Reviews are conducted by presenting not only brief introduction of each method and processing operations on the sunspot number data, but also compare each one in aspects of outcomes, predictibilities and conclusions. The result of this review is that all 5 methods covered here have strength and limitations, but predictions are similar.

Keywords—Machine learning application, Sunspot prediction, Space weather, Machine learning algorithms, solar cycle

I. INTRODUCTION

Sunspots are local sinks on the Sun's photosphere that appears seems to be less bright than other regions on the sun. They are zones of lessened surface temperature achieved by groupings of appealing field movement that frustrate convection, whose number changed in a certain period of time of sunlight [1]. In addition, the influence of the sun's activity will directly affect many aspects of the modern society includes navigation, radars and so on. It also directly or indirectly affects the Earth's climate and organisms [2, 3]. For instance, El Nino-southern oscillation (ENSO), which is an important factor of the earth's climate, has long-term correlation with sunspot cycles. The ENSO cycles influence on the variation of the climate parameter in surrounding countries and regions climate, which depends on the changes in solar activity [4].

Because of the development of big data resources, there has been a rapid growth of use of artificial intelligence (AI) to utilize the data for either prediction or analyzing purposes in astronomy [5]. In particular, machine learning models have been used to conduct space weather research about sunspot prediction [6-8]. There are many existing papers with introductions to utilize machine learning techniques, which will be addressed in later sections in detail. Moreover, supervised learning is a main way to tackle the problem of regression and clustering, which is characterized by labeling the given feature to train a promising model for future prediction.

The structure of the essay is shown as below: In section 2, methodologies of prediction are reviewed in details as well as corresponding models. In section 3, conclusions of different models are addressed and compared. Finally, a summary will present as the result of this literature review.

This topic remains important for space weather prediction and for safety of satellites in space. For example, minimum solar activity could suggest relatively safe space conditions for working astronauts in absence of the Earth's atmosphere.

II. METHODS

In most cases, assumption of the existence of long-term modulation of sunspot number was made by researchers, in which is the basic hypothesis of almost every model. Due to the non-linear fact of dynamics of solar cycle activity, many sophisticated methods were devised to reconstruct sunspot number variable in the time series [9]. There exist some review papers in the literature with more accurate explanations for previously methods [10], but here describes novel ones which predict sunspot number in the near future.

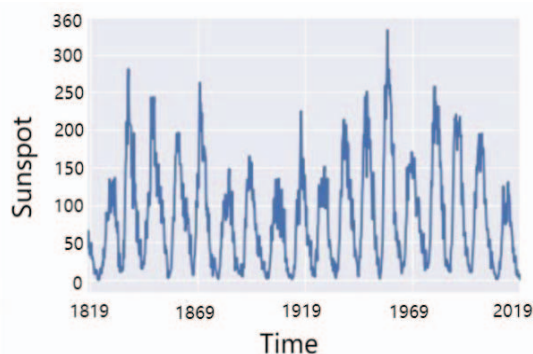


Fig. 1. The number of observed sunspots from 1819 to 2019 (Khan 2020)

In the case of this topic, sunspot observation data is commonly chosen from World Data Center's newly constructed annual sunspot time series, Version 2.0 [6]. Fig.1 shows one example of this data.

A. Bayesian machine learning model

Bayesian machine learning model devices both supervised learning and unsupervised learning [7]. This methodology to predict next solar Cycles by combining algorithms of Machine Learning and iteration steps. Fundamentally, Least-Squares Support-Vector Machines (LS-SVM) regression is incorporated in each iteration.

Specifically, the Non-linear Auto Regressive Exogenous (NARX) model was used to create solar cycles hindcast and forecast model. This model uses past output into the new input to generate a sequence. It can be expressed as $y = f\{y(k-1), y(k-2), \dots, y(k-p), u(k-1), u(k-2), \dots, u(k-p)\}$, p is

the predicted SSN at time k , y stands for the output data and u means the input data, f is modeled by the sum of weighted input data u_k at time k plus a bias term. Fig. 2 shows algorithm for this model.

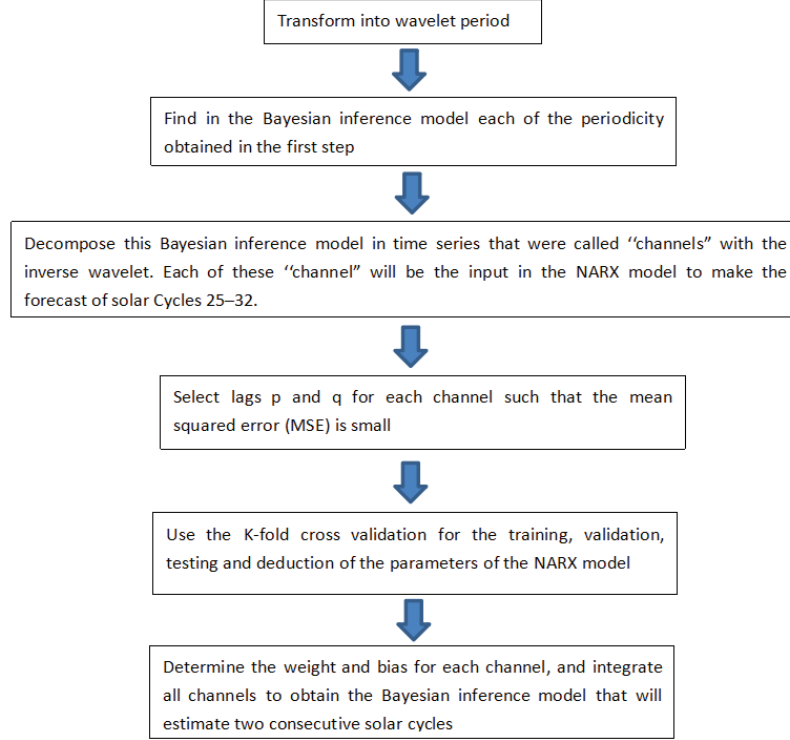


Fig. 2. Flow chart of main algorithms of Bayesian machine learning model

B. Empirical model decomposition

The deep learning long short-term memory (LSTM) model is used to estimate the oscillatory components. The Vanilla Recurrent Neural Network (RNN) model includes state as follows: $ht = \tanh(Wxt + Wht - 1 + bih)$, where Wx is the weights term and bi is the bias term. It uses multiplicative gate units to allow the respective cell to pass or destroy the data [1, 6]. It also has three gates, input, output, and forget (Table I), which estimate the activation of a biased total of input data and hidden states previously using a sigmoid function. So, it indicates the number of allowed relevant information that flows in. The three gates are represented as follows, where $\sigma(s)$ is the sigmoid function. In order to regulate the information flow, the LSTM adds Cell state with above three gates [8].

TABLE I. INPUT(IT), OUTPUT(OT) AND FORGET(FT) GATES

$it = \sigma(ixt + iht - 1 + bi)$
$ot = \sigma(oxt + oht - 1 + bo)$
$ft = \sigma(fxt + fht - 1 + bf)$

Empirical mode decomposition (EMD) designed to extract oscillatory components from time series with different frequencies and convert the oscillatory components into Intrinsic Mode Functions (IMFs) satisfying a set of conditions. It can decompose a target time series into a finite number of IMF by a sifting process in EMD. Generally, $n = \text{ceil}(\log_2 N)$, where n is the number of IMFs and N is the record length as show in Fig 3 [8].

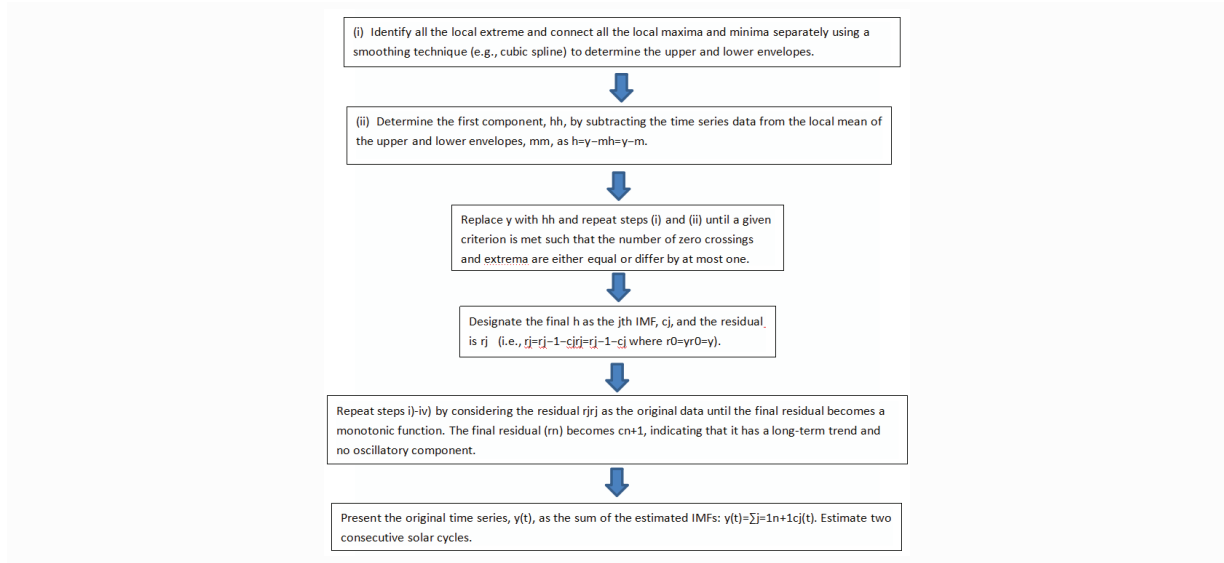


Fig. 3. Flow chart of main algorithms of Empirical model decomposition model

Note that the lower-order components have higher frequency levels and vice versa. For example, the component with the highest frequency level is c_1 , and the component with the lowest frequency level is c_n , while c_{n+1} is the long-term trend without oscillation [8].

C. Neural network autoregression (NNAR), neural network autoregression(R)

The NNAR model can be used in the R environment as a parametric and nonlinear estimation model with the parameters p and k . NNETAR (p, k) is a neural network predicting a single hidden layer, where p is the lag input and k is the number of nodes in the hidden layer [1, 6]. Below Fig. 4 shows the algorithm.

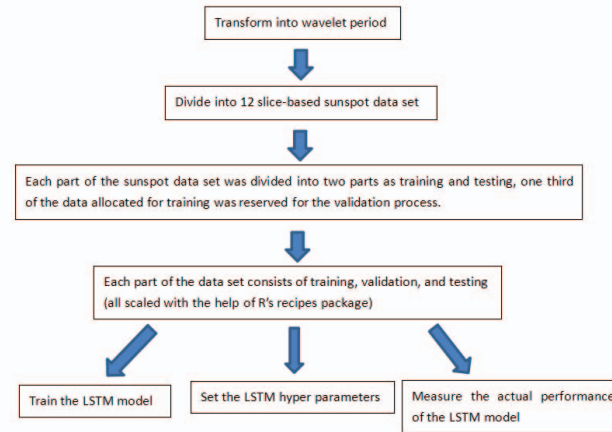


Fig. 4. Flow chart of main algorithms of neural network auto-regression

The data required for training is large, and only one training based on the existing data may not get the expected results for performance. The study also compared other non-ML models, such as ARIMA. But they are not suitable for SSN prediction as suggested.

D. Long Short-Term Memory

This research uses long short-term memory (LSTM), which is a subset of Recurrent Neural Network (RNN).

After training and testing the data, LSTM then predict the next period of SSN without slicing dataset [1].

E. Gaussian Process

Even though a number of methods based on NN have applied in this field, Gaussian Process (GP) has seen its own performance. Compared with other methods, it is also suitable for small data sets, and the output has confidence intervals [1].

In general, this model is composed of a mean function and a kernel function, $f(x) \sim GP(m(x), k(x, x_0))$. However, the error function(k) was represented by a normal distribution in practice. Joint Gaussian distribution is formulated with joint distribution of training and testing points. Warping function is employed to filter out negative values [6].

For training, it aims to maximize log-likelihood which contains a negative covariance matrix Kmm and parameters P, Q. So, obtaining optimized parameters requires training by a genetic algorithm where fitness functions are evaluated to be optimized [6].

III. DISCUSSION

According to the different modeling, the conclusions of each prediction scheme are (with order of section 2):

- The ML model begins with sunspot cycle 24 (circa 2008-2019), and a new extended solar minimum phase will continue until sunspot cycle 27 (around 2050). It is expected that the long-term sunspot minima in the 21st century could be similar to the Dalton minima. Meanwhile, it predicts a peak 95 for Cycle 25 around 2024 [7].
- It will be similar to that of Solar Cycle 24 with peak around 2025 [8]. In this study, the magnitude of the Cycle 25 is expected to be greater than the previous one [2].
- Mostly it can be predicted that cycle 25 will be reached between 2021 and 2025[1].
- According to the results, the peak of Cycle 25 is most likely to occur in 2024. Considering that a median of 110 is the predicted maximum, it can be concluded that Cycle 25 is expected to be slightly smaller than both Cycle 24 and 23. Cycle 25 is likely lower than most modern solar activity, which can have profound implications for the magneto-ionosphere [6].

In summary, the prediction results are shown in the Table II.

TABLE II. PREDICTION RESULTS

Model	Result
2.1 Bayesian machine learning model	Peak annual around 2024, 95+-(100).
2.2 EMD-LSTM model	Peak in 2025, smaller than cycle 23, larger than cycle 24.
2.3 NNAR, neural network autoregression(R)	It will have greater value than the previous.
2.4 LSTM	Cycle 25 will between 2021 to 2025
2.5 Gaussian Process	Peak in 2024 with 100-127, smaller than previous.

In comparison, NNAR, LSTM and Bayesian modeling are illustrated as high-confidence interval outputs. Although complex algorithms are needed, their performance is quite accurate compared with other algorithms.

IV. CONCLUSION

For most modeling, the result is that Cycle 25 is expected to occur around 2025. The magnitude is moderate compared to before, followed by a minimum near 2030.

However, since solar activity is a highly complex and variable system. and ML model estimates are limited by its uncertainty principle, it can only predict the probability ranges of the average sunspots number in a given year [7]. The quality of its estimated forecast parameters is limited by the uncertainty principle [7]. This paper purely reviews the methods in theories, but did not focus on efficiency and practices of performance of the models due to lack of operational guidance. To improve, further details should be added on aspects of analysis of performance and realization of the models discussed above. Future innovative directions for this topic would be verifying predictabilities of those models with actual observation data, or either devise new method for prediction of long-term meaning of sunspots.

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